

Kids Piano — Implementation Spec v4 (final)

Status: supersedes `KidsPiano-app-and-screens.pdf`, `Kids_Piano_App_Master_SPEC_FINAL.md` and `gemini-code-1789815228334.md` wherever they disagree. Written 19 September 2026.

Audience for this file: the developer or AI coding agent who will build releases 0.5.0, 0.6.0 and 0.7.0.

0. How to use this document

1. The installed app is **0.4.0** (package `com.vijaychhetry.kidspiano`): one practice screen, Calibrate, Audio Lab. Before writing code, read the existing code and compare it with §4–§6. Keep every existing test. Where existing behaviour conflicts with this spec, follow this spec and note it in the PR.
 2. Build in release order (§14). A release ships only when its gate is met.
 3. Every number marked **(tunable)** is a starting value. Put all of them in one `Config` object (§4.1) so they can change without touching logic.
 4. Three product assumptions are still open (§1.2). Their defaults are used throughout; each is one constant or one composable to change.
 5. Nothing in this file requires a network, an account, a cloud service or analytics. Do not add any.
-

1. Product

1.1 What it is

An Android app that listens to a real piano (tested on a Yamaha PSR-F52) through the phone microphone and teaches a young child five white keys, one at a time: **C4, D4, E4, F4, G4**. There is no cable and no MIDI. The phone names the note that was played and answers.

Nothing outside C4–G4 is ever asked for, in any level, song or game.

1.2 Assumptions awaiting the owner's confirmation

ID	Assumption (default used)	Where it bites
A1	Target age is 6–8, designed at 7 . Short words plus icons; letters and colours carry meaning.	Copy length, gate difficulty, session length
A2	A session is three activities, about 8 minutes : Warm-up → Learn → Play.	§8, §10
A3	The Grown-ups gate is a times-table question plus a 2-second press-and-hold .	§9.9

1.3 Non-negotiable rules

These each get their own automated test (§13).

1. **Audio the app cannot understand is never called wrong.** Two notes, a pitch between keys, noise, too quiet, too short, or an unsure octave all give a neutral message. Never a cross, a lost star or a miss.
2. **One physical press produces exactly one judgement.** A key that stays held is never judged twice.
3. **A press cannot answer a prompt that appeared after the press started.**
4. **No level can be failed.** No lives, no timers, no game-over, no locked door for quitting early.

5. **No audio is ever recorded, stored or sent.** Sound is analysed as it arrives and discarded. Disk holds only numbers. The release build contains no recording or file-write code for audio.
6. **The app plays no sound.** The piano is the only instrument, and anything the phone plays would be heard by its own microphone.
7. **Colour is never the only signal.** Every colour is accompanied by the note letter.
8. **No mascot, streaks, coins, shop, accounts, cloud or analytics.**

1.4 Not building (do not add)

Character system, Piano World map, sticker room, parent dashboard beyond the Progress tab (§9.12), Piano Train, Pop the Note, Star Catcher, sheet music, black-key lessons, chords, two-hand playing, notes outside C4–G4, the app playing songs aloud, Room, Hilt, cloud backup.

2. Fixed vocabulary

2.1 Keys and colours

Note	MIDI	Hz (A4=440)	Colour	Hex	Text on colour
C4 (middle C)	60	261.63	Red	#E5484D	ink
D4	62	293.66	Orange	#F2731E	ink
E4	64	329.63	Yellow	#F1B92B	ink
F4	65	349.23	Green	#2FA66A	white
G4	67	392.00	Blue	#3B82E0	white

Use the same colour for the same key on the keyboard guide, mini-map highlight, hero note, song bubbles, falling lanes and the My Five Keys strip. Tinted state = colour mixed 26% into white.

Always write "**C4 (middle C)**" or "**middle C**" next to C4 in setup, on stickers and in Audio Lab. Yamaha manuals call middle C "C3"; the app uses scientific pitch (C4).

Lesson sets: **C4–G4** (default) and **C3–G3** (for a piano whose upper register is hard for the phone). Everything in this spec is written for C4–G4; the lower set is the same logic shifted down an octave, with MIDI 48, 50, 52, 53, 55.

2.2 Design tokens

Token	Value
primary	#5B3CC4
background	#FFF7F0
surface	#FFFFFF , 1.5 dp border #EADFD6
ink	#27232B
inkMuted	#756E7E
success	#2FA66A
amber (wrong key ring, unclear pulse)	#E7A34B
disabled	#D8D3DC
Piano bed (keyboard background)	#221E2D

Never use red for feedback. Red is only ever the colour of C4.

Typography: **Nunito**, bundled as a font resource (works offline).

Role	Size	Weight
Hero note	88–96 sp	900
Screen title	24–28 sp	900
Feedback sentence	16–18 sp	900
Body	16 sp	700
Helper	14 sp	700
Grown-ups link	14 sp	800

No essential instruction below 14 sp. Child buttons are at least 56 dp high, 16–20 dp radius, 16 dp screen margins. The primary action sits in the same place on every screen. Support system font scaling to 1.3× without clipping the primary action; the keyboard guide is a drawing and does not scale with text.

2.3 Motion and haptics

Event	Effect
Sound onset	Listening ripple on the badge within ~50 ms
Correct	200 ms key glow in its colour, small star pop
Wrong key	Amber ring on the pressed key. No shake, no red. Ring persists until the next prompt (see §9.3)
Unclear	Soft pulse on the badge. No ring, no arrow, no cross
New key introduced	500 ms spotlight on the key
Screen transition	200 ms slide; forward slides left, back slides right
Reward reveal	600–800 ms stars scale-in

Respect the system "remove animations" setting. **Haptics are off by default** (a phone on a music stand can pick up its own buzz through the microphone). Add a grown-up toggle, and verify in Audio Lab that haptics do not raise the signal level before enabling.

3. Architecture

Keep the existing app structure where it works. Suggested packages under `com.vijaychhetry.kidspiano`:

```
audio/      AudioInput (AudioRecord wrapper),
SourceSelector
pitch/      PitchDetector interface + Yin/Mpm
implementation, FundamentalCheck
press/      OnsetDetector, PressTracker (state machine),
PressEvent
calibration/ CalibrationEngine, CalibrationProfile,
Classifier
judge/      Judge (expected vs press → Verdict),
Verdict
curriculum/ Level, LevelPlan (prompt generator), Song,
SongLibrary, Stars, Planner
session/    SessionEngine (activities), HintLadder,
SessionLogger
storage/    ProgressStore, CalibrationStore, LogWriter,
Exporter
ui/theme/   tokens, Nunito
ui/components/ NoteHero, KeyboardGuide, MiniMap,
PromptPips, Badge, FeedbackLine, FiveKeys, StopRow,
SongBubbles
ui/screens/ Home, LevelMap, Drill, Song, Summary,
FreePlay, Falling (0.7), Memory (0.7),
GrownUps (Gate, Setup, Calibrate, AudioLab,
Files, Progress)
```

Stack: Kotlin, Jetpack Compose, Material 3, Navigation Compose, Coroutines and Flow, existing preferences plus JSONL files. **The UI**

never touches pitch-detection code; it consumes `PressEvent` and `Verdict` only.

Core interfaces:

```
interface AudioInput { fun start(): GrantedFormat; fun
stop(); val frames: Flow<AudioFrame> }
interface PitchDetector { fun detect(window: FloatArray,
sampleRate: Int): PitchResult }
data class PitchResult(val hz: Double?, val clarity:
Double, val confidence: Double)

data class PressEvent(
    val onsetNanos: Long,          // CLOCK_MONOTONIC, from
audio sample position
    val verdictNanos: Long,        // when the judgement was
decided
    val kind: PressKind,          // NOTE, TWO_NOTES,
NOISY, LOW_CONFIDENCE, TOO_SHORT, TOO_QUIET
    val hz: Double?, val midi: Int?, val centsOff:
Double?,
    val confidence: Double, val clarity: Double, val
levelDb: Double,
    val octaveEvidence: OctaveEvidence?
)

sealed interface Verdict {
    data class Correct(val press: PressEvent): Verdict
    data class WrongKey(val heard: Note, val press:
PressEvent): Verdict
    data class WrongOctave(val heard: Note, val press:
PressEvent): Verdict
    data class Unclear(val reason: UnclearReason, val
press: PressEvent?): Verdict // never counts
}
enum class UnclearReason { TWO_NOTES, NOISY,
LOW_CONFIDENCE, OCTAVE_UNSURE, TOO_QUIET, TOO_SHORT,
NO_SIGNAL }
```

4. Audio engine

4.1 Config (all tunable)

```
object Config {
  // capture
  val preferredRates = listOf(48_000, 44_100)
  const val WINDOW = 2048
  const val HOP = 512
  // onset
  const val ONSET_RISE_DB = 6.0           // rise over
the last 2 hops
  const val ONSET_MIN_ABOVE_FLOOR_DB = 12.0
  const val MIN_ONSET_GAP_MS = 150
  const val ATTACK_SKIP_MS = 20          // ignore
pitch windows that include the attack transient
  // pitch
  const val F_MIN = 120.0                 // ≈ C3 minus
margin
  const val F_MAX = 1100.0               // ≈ C6 plus
margin
  const val MIN_CLARITY = 0.85
  const val MIN_CONFIDENCE = 0.80
  const val AGREE_FRAMES = 3
  const val CLASSIFY_MAX_CENTS = 50.0
  const val TWO_NOTE_SALIENCE_DB = 10.0  // second
non-harmonic peak within this of the first
  const val OCTAVE_FUNDAMENTAL_DB = 6.0
  // release / timeouts
  const val RELEASE_BELOW_PEAK_DB = 20.0
  const val PRESS_TIMEOUT_MS = 350       // no stable
agreement by then → unclear/too short
  const val MIN_LEVEL_DB_ABOVE_FLOOR = 10.0
  // idle
  const val HINT_IDLE_MS = 4_000
  const val HINT_SILENCE_AFTER_WRONG_MS = 12_000
  const val PROMPT_NUDGE_MS = 20_000
  const val MIC_RELEASE_MS = 60_000
  // planner
  const val COMFORT_FIRST_TRY = 0.80
```



```
const val COMFORT_MIN_SAMPLES = 6
}
```

4.2 Capture

- Use `AudioRecord`, mono, PCM 16-bit.
- **Sample rate:** try 48 000 Hz, then 44 100 Hz, on `AudioRecord` itself and use the rate it actually grants. **Do not use** `AudioManager.PROPERTY_OUTPUT_SAMPLE_RATE`; that is the output rate.
- **Source fallback chain:** `UNPROCESSED` (only if `AudioManager.PROPERTY_SUPPORT_AUDIO_SOURCE_UNPROCESSED` is "true") → `VOICE_RECOGNITION` → `MIC`. On whichever is used, disable `AutomaticGainControl`, `NoiseSuppressor` and `AcousticEchoCanceller` where the device allows. Report the chosen source and rate in Audio Lab.
- Read on a dedicated thread at `Process.THREAD_PRIORITY_AUDIO`. Use a buffer of at least $2 \times \text{getMinBufferSize}$.
- Capture the frame-position → monotonic-time mapping with `AudioRecord.getTimestamp(ts, AudioTimestamp.TIMEBASE_MONOTONIC)`. If it is unavailable, estimate from `System.nanoTime()` at read time minus the buffer duration and flag `timestampEstimated = true` in the log.
- Microphone is open **only** while a listening screen is visible. Release it on pause, phone calls, focus loss, screen off, and after `MIC_RELEASE_MS` of silence (show "Tap to keep playing"). Keep the screen on (`FLAG_KEEP_SCREEN_ON`) while listening.
- Request `RECORD_AUDIO` with the real system dialog. If denied, show the permission screen (§9.14).

4.3 Onset detector

Runs on every hop (512 samples), independent of pitch.

```
rmsDb[t]   = 20·log10(rms of the hop)
floorDb    = slow running minimum/percentile of rmsDb
during silence
```

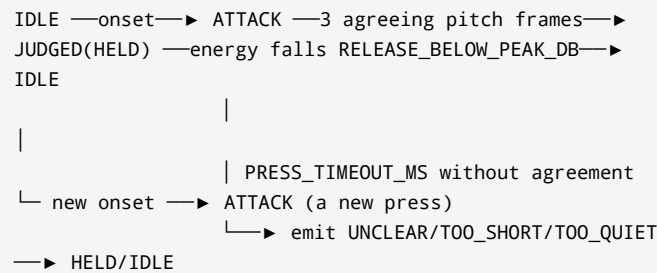
```

onset(t)  = rmsDb[t] - rmsDb[t-2] >= ONSET_RISE_DB
           && rmsDb[t] >= floorDb +
ONSET_MIN_ABOVE_FLOOR_DB
           && (now - lastOnset) >= MIN_ONSET_GAP_MS

```

The first hop that satisfies this defines `onsetNanos`, computed from the sample index (refine to the hop start; sub-hop accuracy is not needed). This is what makes a **re-press of the same key** detectable while the previous note is still decaying: the energy jumps again. It is also what stops a held key from being judged twice: no new jump, no new press.

4.4 Press tracker (state machine)



Rules:

- **Exactly one `PressEvent` is emitted per onset.** After it is emitted the tracker ignores pitch until the next onset or a release.
- In `ATTACK`, skip pitch windows whose start is earlier than `onsetTime + ATTACK_SKIP_MS`.
- Use overlapping windows: window 2048, hop 512. `AGREE_FRAMES` = 3 consecutive windows whose detected note is the same and whose `clarity` $\geq$ `MIN_CLARITY`. Expected onset-to-verdict is about 70–100 ms; do not claim a number until measured (§13.3).
- **Classify** by nearest calibrated key in log-frequency (§4.6). Accept only if within `CLASSIFY_MAX_CENTS` of the nearest note and clearly closer to it than to the next.

- **Two notes:** if a second, non-harmonically-related spectral candidate is within `TWO_NOTE_SALIENCE_DB` of the strongest → `TWO_NOTES` .
- **Too quiet:** peak level less than `MIN_LEVEL_DB_ABOVE_FLOOR` above the floor → `TOO_QUIET` .
- **Too short:** energy fell below the release threshold before three agreeing frames existed, but the peak was above the floor → `TOO_SHORT` .
- **Otherwise on timeout** → `LOW_CONFIDENCE` or `NOISY` .
- **Octave errors:** pitch trackers jump octaves on piano tones. Compute Goertzel (or a narrow FFT bin) energy at the expected note's fundamental and at the detected note's fundamental over the agreed frames.
 - Expected note E, detected note D is $E \pm 1$ octave: if the fundamental of E carries energy within `OCTAVE_FUNDAMENTAL_DB` of D's (or more), treat the press as **E** (correct).
 - Report `WRONG_OCTAVE` only when the three frames agree on D **and** D's fundamental is clearly present **and** E's fundamental is clearly absent. Otherwise `OCTAVE_UNSURE` (unclear).
 - This check needs the expected note, so it lives in `Judge` , not in the tracker; the tracker supplies `octaveEvidence` .
- Black-key presses (a note not in the lesson set) are classified as `WrongKey(heard = C#4 etc.)` only with high confidence; otherwise unclear.

4.5 Frequency and note maths

```

midi = 69 + 12·log2(f / 440)
cents(f, ref) = 1200·log2(f / ref)

```

Convert to a nearest note only after clarity and confidence checks. The detector search range is `F_MIN..F_MAX` (about C3–C6): wide enough for the C3–G3 lesson set and for spotting a wrong octave, narrow enough to remove most subharmonic errors. Use YIN or MPM

(TarsosDSP is acceptable) behind `PitchDetector` . Do not use a loudest-FFT-peak detector.

4.6 Calibration

- **Range:** white keys A2–F6, the range the detector can actually hear. The mini-map fades the rest.
- **Procedure per key:** the grown-up plays the named key four times (`sample n of 4`). Reject samples with low clarity, two notes or level below threshold. Store per key: median Hz, spread in cents, sample count, mean confidence.
- **Key dropdown** at the top: picks any key in range and jumps to it. Re-measuring a key **clears its samples and starts again**; old and new readings are never mixed.
- **Profile:** `quality` in {EXCELLENT, GOOD, FAIR, POOR} from spread and confidence. Store also `sampleRate` , `micSource` , `pianoName` (optional free text), `createdAt` .
- **Use:** classification compares detected Hz with each calibrated key's median. Ignore an overall tuning offset below about 25 cents. An uncalibrated key falls back to equal temperament corrected by the mean offset of the calibrated ones.
- **Listen test** (last setup step): the grown-up presses each of the five keys once; each must be named correctly. Pass is 5 of 5.
- Recalibrating overwrites the profile after confirmation ("Replace the piano setup?").

4.7 Judge

```
fun judge(expected: Note, press: PressEvent,  
promptShownAtNanos: Long): Verdict?
```

1. If `press.onsetNanos < promptShownAtNanos` → return `null` (rule 1.3.3: the press began before this prompt, so it cannot answer it). It is still logged.
2. Map non-NOTE kinds to `Unclear(reason)` .

3. NOTE kind: apply the octave logic (§4.4). Same MIDI as expected
→ `Correct` . Same pitch class, other octave, high confidence →
`WrongOctave` . Otherwise → `WrongKey` .
4. Never return `WrongKey` when confidence or clarity is below
threshold; return `Unclear(LOW_CONFIDENCE)` .
-

5. Data and files

5.1 Progress store (preferences, JSON string)

```
{
  "v": 1,
  "stops": { "1": {"bestStars": 3, "lastPlayedAt":
1789800000000}, "...": {} },
  "keys": { "60": {"firstTry":
[true,true,false,true,true,true]}, "62": {"firstTry": []}
},
  "songs": { "hot_cross_buns": {"plays": 2, "bestStars":
3} },
  "repeatOffered": { "3": true },
  "settings": { "haptics": false, "lessonSet": "C4-G4" }
}
```

`keys[...].firstTry` keeps the last 30 press outcomes per key from
Learn activities (first-try correct or not, helped counts as not first-try).
`totalStars` = sum of best stars over stops and songs.

5.2 Session log (`session-YYYY-MM-DD.jsonl` , one line per judged press, append only)

```
{
  "v": 1, "ts": 1789800000123, "sessionId": "s-...", "appVersion": "
0.6.0", "activity": "learn",

  "levelId": 4, "promptIndex": 3, "expected": "C4", "expectedMidi"
: 60,

  "heard": "E4", "heardMidi": 64, "verdict": "WRONG_KEY", "unclear
```

```
Reason":null,

"confidence":0.94,"clarity":0.97,"levelDb":-31.2,"centsOff":-4.1,

"onsetNanos":123456789000,"verdictNanos":123553000000,"onsetToVerdictMs":96,

"timestampEstimated":false,"hinted":false,"helped":false,"wrongCountThisPrompt":1}
```

Presses discarded by rule 1.3.3 are logged with
`"verdict":"IGNORED_STALE"` . No audio, no samples.

5.3 Calibration export (`calibration-YYYY-MM-DD.json`)

```
{ "v":1,"createdAt":"...", "quality":"EXCELLENT", "sampleRate":48000, "micSource":"UNPROCESSED",
  "pianoName":"PSR-F52", "keys":
  [{ "note":"C4", "midi":60, "expectedHz":261.63,

    "medianHz":262.1, "spreadCents":3.2, "samples":4, "confidence":0.97} ] }
```

Live profile stays in private prefs (`shared_prefs/calibration.xml`); it is unreadable without root, which is why export exists. Both exports go through the Android share sheet via `FileProvider` .

5.4 Printable stickers

Grown-ups → Setup → "Print stickers" generates an A4 PDF with `PdfDocument` , offline: five circles per set (Ø 14 mm), each in the key's colour with its letter and note name, C labelled "C4 · middle C". Three sets per page. Share/print via the system sheet.

6. Curriculum

6.1 Ladder (eight stops)

#	Stop	Keys	Learn-phase prompts	Structure
1	One key: C	C	6	C ×6 (repeats allowed; only one key)
2	Two keys: C and D	C D	6	C D C D C D
3	Mix it up	C D	8	Shuffled, 4 each
4	Three keys: E joins	C D E	8	C D E in order, then 5 shuffled (E at least twice)
5	Four keys: F joins	C D E F	9	In order (4), then 5 shuffled (F at least twice)
6	All five: G joins	C D E F G	11	In order (5), then 6 shuffled (G at least twice)
7	Song time	learned song	song length	A real tune at the child's own pace
8	Falling notes	C–G	12 tiles	The game (0.7.0)

A level is capped at 11 prompts.

Shuffle rules (all shuffled parts): balanced, so each key appears about the same number of times; **never the same key three times in a row**, checked across the ordered/shuffled boundary; the newly joined key appears at least twice. Stop 1 is exempt from the "three in a row" rule.

Unlock: a stop unlocks when the stop before it earns at least one star. Stops 7 and 8 use the same rule. Songs are separately available as soon as their keys are learned (§7).

6.2 Stars (Learn phase only; warm-up never counts)

Let n = prompts in the level, $wrong$ = prompts that had at least one **WrongKey/WrongOctave** verdict. Unclear never counts. A **helped** prompt (§8.3) is not counted as wrong.

Stars	Rule
★★★	$wrong \leq \max(1, \text{floor}(0.15 \cdot n))$ and every prompt completed
★★	$wrong \leq \text{floor}(0.40 \cdot n)$ and every prompt completed
★	at least 80% of prompts completed, even if the child stops early

A child who plays nothing earns nothing (no prompts completed). Stopping early with $\geq 80\%$ completed still shows the summary and banks a star. Quitting is never punished with a locked door.

6.3 Falling notes stars (0.7.0)

Stars from **catches** only: ★ ≥ 6 of 12, ★★ ≥ 9 , ★★★ ≥ 11 (tunable). No miss tally is ever shown.

6.4 Planner: "Today's adventure"

Runs when Home is opened.

1. **next** = the first unlocked stop with 0 stars; if none, the lowest stop with fewer than 3 stars; if none, Free Play.
2. **Comfort:** a key is *comfortable* when its last two Learn sessions have at least `COMFORT_MIN_SAMPLES` presses and first-try rate $\geq$ `COMFORT_FIRST_TRY`. Keys with too little data are *unknown*, not weak.

3. If the stop before `next` has ≥ 1 star and one of its keys is *weak* (enough data, below threshold), recommend **repeating that earlier stop once** (`repeatOffered[stop] = true`), labelled "Again: Level 3". Never offer a repeat twice for the same stop.
4. Warm-up keys = comfortable keys among the learned ones; if none, all learned keys. Never include the key that this stop introduces.
5. The map still lets the child open any unlocked stop. The planner only chooses what the Home button plays.

7. Songs

All are traditional or original, every note within C4–G4, no rhythm requirement: **the strip waits for the child**. A wrong note does not advance and does not end anything; it gets the same answer it would in a drill.

Offered on the Summary as soon as every note is a learned key.
Sequence = the notes to play; syllables shown under each bubble.

id	Title	Needs	Notes	After	Sequence (note · syllable)
<code>step_up</code>	Step Up (warm-up)	C D E F	8	Stop 5	C·Up D·up E·up F·up F·down E·down D·down C·down
<code>au_clair</code>	Au Clair de la Lune	C D E	11	Stop 4	C·Au C·clair C·de D·la E·lu D·ne C·mon E·a D·mi

id	Title	Needs	Notes	After	Sequence (note · syllable)
					D·Pier C·rot
hot_cross_buns	Hot Cross Buns	C D E	17	Stop 4	E·Hot D·cross C·buns E·Hot D·cross C·buns C·One C·a C·pen D·ny D·two D·a D·pen D·ny E·Hot D·cross C·buns
are_you_sleeping	Are You Sleeping (first half)	C-G	14	Stop 6	C·Are D·you E·sleep C·ing C·Are D·you E·sleep C·ing E·Broth F·er G·John E·Broth F·er G·John

id	Title	Needs	Notes	After	Sequence (note · syllable)
ode_to_joy	Ode to Joy (first line)	C-G	15	Stop 6	E E F G G F E D C C D E E D D (syllables: la)
mary_lamb	Mary Had a Little Lamb	C D E G	26	Stop 6	E·Ma D·ry C·had D·a E·lit E·tle E·lamb D·lit D·tle D·lamb E·lit G·tle G·lamb E·Ma D·ry C·had D·a E·lit E·tle E·lamb E·its D·fleece D·was E·white D·as C·snow

Note: "Are You Sleeping" and "Ode to Joy" are deliberately partial because their next phrases need A. Each stops on a complete musical phrase.

Prominence order on the Summary: newest-unlocked first, then by shortest length.

8. Sessions

8.1 Shape (assumption A2)

Every stop launches a session of three activities, about 8 minutes:

Activity	What	Counts for stars
Warm-up	~6 prompts from the comfortable, already-learned keys, shuffled. Skipped on stop 1. Never introduces the stop's new key.	No
Learn	The stop's drill (§6.1).	Yes
Play	Offered from the Summary: a song if one is unlocked (newest highlighted), otherwise Free Play for about 60 s. Optional.	Songs earn stars per §6.2 with prompts = notes

"Done for today" is visible on every screen after the Warm-up and is one tap. "One more" replays the level or opens the offered song. Ending early saves everything already earned. Confirm leaving with a two-button dialog: **Keep playing** / **End session**.

Stops 7 and 8 are single-activity: Song time is the song picker, Falling notes is the game.

8.2 Prompt lifecycle

```
show prompt (record promptShownAtNanos)
  → wait for PressEvent
  → Judge(expected, press, promptShownAtNanos)
    Correct      → success feedback → next prompt (same
key twice: "Yes! C4 again.")
    WrongKey     → amber ring, sentence, stay on prompt,
wrongCount++
    WrongOctave  → amber ring on octave key, sentence,
stay, wrongCount++
    Unclear      → neutral pulse, stay, no counter
change
    null (stale) → ignore
```

A prompt is complete on Correct or Helped.

8.3 Hint ladder (never trap the child)

Trigger	Effect
4 s with no press	Target key on the keyboard pulses softly
2nd WrongKey/WrongOctave on this prompt	Target key ringed purple; distance sentence shown
3rd wrong, or 12 s of silence after a wrong	"Play the glowing key." Pressing it completes the prompt as helped : no star lost, not counted as first-try in Progress
20 s of complete silence	"Still there? Press any key."
60 s of complete silence	Listening pauses, mic released, "Tap to keep playing."
3 unclear results in a row	Small grown-up hint: "Try moving the phone closer to the speaker."

9. Screens

All child screens are portrait, 360 × 800 dp baseline; test 360×800, 390×844, 412×915 and font scale 1.3×. Full-screen activities hide any bottom navigation. Android back on any activity shows the leave dialog (§8.1).

9.1 Home (child)

- Wordmark, five colour dots.
- Card "Today's adventure": stop name, "3 short games · about 8 minutes".

- ► **Play** (64 dp) runs the planner recommendation. Disabled until calibration exists; tapping disabled Play opens the modal "Ask a grown-up to help wake up the piano first!" with a shortcut to the gate.
- **My Five Keys:** five key shapes; learned keys filled with their colour, others dashed grey. Star total beside it.
- Status chip: green "Piano is ready" (with "Calibrated today · 5 keys · good") or the grown-up message.
- "Grown-ups": small text link at the bottom, deliberately unglamorous; **press-and-hold 2 s** to open the gate.
- Nothing else. No streaks, no "come back tomorrow".

9.2 Level map (child)

- "Pick a stop". Stops in a zig-zag path: number, name, key dots, up to three stars. Current stop outlined purple; locked stops dimmed with a padlock and not tappable. Stops 7 and 8 use gold badges (♪ and ★). In 0.6.0 stop 8 reads "Falling notes · soon".
- Scrolls to the current stop on open. A bottom **Play Level N** button is the same action as Home's Play.

9.3 Drill (child): Warm-up and Learn

Layout, top to bottom: ◀ Map and stop name; "Play this key"; **NoteHero** in the key's colour (C4); caption "the red key · middle C" (middle C only for C4); **Badge**; **FeedbackLine**; **PromptPips**; **MiniMap** (all 61 keys C2–C7, thin bars, the target key highlighted, caption "where it is on the whole piano"); **KeyboardGuide** (C4 to C5, at least 110 dp tall).

KeyboardGuide: target filled solid; keys taught in this stop tinted; the pressed key ringed amber; labels on C4–G4 and C5. The **amber ring persists until the next prompt appears**, not for the ~200 ms the tracker remembers a press.

State	Badge	Line 1	Line 2
Waiting	♪ listening pulse	—	—
Any sound starts (~50 ms)	ripple	—	—
Correct	✓ green	Yes! Now D4. (or Yes! C4 again.)	—
Wrong key	⌂ amber	That was E4 .	C4 is two keys to the left ←
Wrong octave	⌂ amber	That was C5.	Press C4.
Unclear	... soft pulse	I couldn't hear that.	One key at a time.
Too short	... soft pulse	I heard something.	Hold the key a bit longer.

The distance sentence uses the count of **white keys** between the pressed and target keys and an arrow; only when the pressed key is a white key inside the visible keyboard. For black keys or others: "That was C#4. Press C4."

Pips: one dot per prompt. Green done, purple current, grey to come. Helped prompts show green.

9.4 Song (child)

◀ Map and "Song · your own pace"; title; "Play the big note. No hurry."; **SongBubbles**: at most six visible, current bubble twice the size and outlined, done bubbles dimmed with a small tick; lyric syllable under each (≥ 12 sp); FeedbackLine; KeyboardGuide (same as drill); "4 of 17". Strip scrolls to keep the current bubble centred. After 4 s idle the target key pulses. No metronome, no timing score. Stars per §6.2.

9.5 Session summary (child)

Stars scale in. Title in plain words: "You know C, D and E!" One line: "11 of 11 on the first try." The just-learned key fills in **My Five Keys**. If songs are unlocked: card "You can play a song now" with the tune buttons, most prominent first (primary style for the first). Then **One more** (secondary) and **Done for today** (text button, always one tap). If the level was stopped early with $\geq 80\%$ completed, this screen still appears and a star is banked.

9.6 Free Play (child, 0.6.0)

KeyboardGuide plus a large area showing the last recognised note in its colour and name. Each correct recognition lights its key. Unclear gives only the soft pulse. There is no "wrong" and no scoring. A Done button leaves. Logged with `activity: "free"`.

9.7 Falling notes (child, 0.7.0)

Five lanes, C D E F G left to right, coloured tiles falling to a line above the keyboard guide. Fall 3 s; spacing 2.5 s; 12 tiles. **Judge on onsetNanos**, **not on verdictNanos**: window is 450 ms early to 900 ms late relative to the tile reaching the line (tunable; deliberately lopsided because beginners are late, not early). A missed tile fades and returns once at the end of the run. A wrong key is named and the tile keeps falling. Unclear is ignored. The counter shows catches only. Stars per §6.3.

9.8 Memory (child, 0.7.0)

The app lights a sequence of keys silently (visual only). The child repeats it. Length starts at 2 and rises to 4 after two successes; six rounds. Wrong key: named, and the sequence replays; nothing ends. No timer.

9.9 Grown-ups gate

Numeric-only question, re-rolled every visit: "What is 7×8 ?" with `a`, `b` $\in [6, 9]$. Wrong answer: "That is not the number." and try again; no

lockout, no scolding. Opened from Home only by press-and-hold of 2 s on the link. This is a speed bump, not security.

9.10 Setup checklist (grown-up, new in 0.5.0)

Landing tab of Grown-ups until calibration exists. Steps with checkboxes and a progress count:

1. Put a sticker on each key (**Print stickers**; note the C is middle C).
2. Set the keyboard to Grand Piano (other voices break pitch and onset detection).
3. Put the phone beside the speaker on a stand (on a PSR-F52 the speaker is on top).
4. Calibrate the five keys (opens §9.11).
5. Listen test: press each key; Audio Lab shows the note.
Primary button continues to the next unfinished step.

9.11 Calibrate (grown-up)

Key dropdown at the top (A2–F6), "Play this key", large note, "sample 2 of 4", keyboard guide with full mini-map (unhearable keys faded), buttons **Start**, **Pause**, **Skip key**, saved-profile line ("EXCELLENT · 5 notes"), **Export calibration**.

9.12 Grown-ups tabs

Tabs: **Setup** · **Calibrate** · **Audio Lab** · **Files** · **Progress**.

- **Audio Lab** (existing, keep as is): live note, Hz, confidence, hint, microphone, sample rate, frames seen/dropped, signal level, **Change mic**, event log with CORRECT / AMBIGUOUS / NO_SIGNAL. Add: chosen audio source, onset events, onset-to-verdict ms, a Haptics test. Help text: frames dropped > 0 means the phone is struggling; AMBIGUOUS usually means a sustain pedal or overlapping keys; NO_SIGNAL means the phone is too far from the piano.
- **Files**: Export calibration; Export session log; **Lesson set** dropdown (C4–G4 default, C3–G3); Haptics toggle.

- **Progress** (new): from the session log and progress store. Per key: first-try %, helped count, unclear rate; minutes per session; last 7 sessions; calibration date. No ranking, no red.

9.13 Leave dialog

"Keep playing" / "End session". Never a scolding tone.

9.14 Permission and edge screens

- **Microphone needed:** "Kids Piano listens to your piano. Nothing is recorded." Buttons Allow microphone / Grown-up help. Uses the real system dialog.
- **No signal:** "I can't hear the piano yet. Move the phone a little closer."
- **Offline:** nothing to show; everything works offline.
- **Calibration required:** redirect to the gate, not a technical error.

10. Copy deck

Use exactly these, or shorter. Avoid: Incorrect, Wrong answer, You failed, Invalid, Score penalty, Recognition failure.

Key	Text
Correct	Yes! Now D4. / Yes! C4 again.
Wrong key	That was E4. + C4 is two keys to the left ←
Wrong octave	That was C5. Press C4.
Unclear	I couldn't hear that. One key at a time.
Too short	I heard something. Hold the key a bit longer.
Nudge	Still there? Press any key.
Mic paused	Tap to keep playing.

Key	Text
Hint	Play the glowing key.
Not calibrated	Ask a grown-up to help wake up the piano first!
Ready	Piano is ready
Summary	You know C, D and E! · 11 of 11 on the first try
Song offer	You can play a song now
Exit	Done for today · One more
Gate wrong	That is not the number.
Song hint	Play the big note. No hurry.

11. Accessibility

Icons plus short words; note names always shown; contrast: ink text on yellow and orange, white on green and blue; 56 dp child buttons; predictable primary-button position; no essential state by colour alone (red and green neighbours are always disambiguated by letter and position); support system font scaling to 1.3x; slower feedback animation when animations are reduced; TalkBack labels on all controls ("Play, button", "Key C4, target").

12. Privacy

Microphone active only during listening screens; local processing only; no network permission (`INTERNET` absent from the manifest); no accounts; no analytics; raw audio discarded as it arrives; stored data are the calibration numbers, progress numbers and the session log. Exports go out only through the user's share sheet.

13. Tests

13.1 Rule tests (each must fail if its feature is broken)

ID	Test
R1	Every <code>Unclear</code> reason returns no stars lost, no wrong count, no miss (property test over all reasons)
R2	A held key produces one <code>PressEvent</code> ; a second is produced only on a new onset
R3	Same key struck twice ~1 s apart (decaying tone + new attack) produces two <code>PressEvent</code> s
R4	A press with <code>onsetNanos < promptShownAtNanos</code> returns null and is logged as <code>IGNORED_STALE</code>
R5	Holding C while the prompt changes to D does not produce a wrong D
R6	No level can be failed: any sequence of inputs leaves the level completable
R7	Playing nothing earns no stars
R8	Stopping at exactly 80% completed banks ★ and shows the Summary
R9	Shuffle: balanced counts; no 3 in a row incl. the ordered/shuffled boundary; new key ≥ 2 times; stop 1 exempt
R10	Unlock: stop N+1 opens exactly when stop N has ≥ 1 star
R11	Wrong octave only when 3 frames agree, D's fundamental present, E's absent; otherwise <code>OCTAVE_UNSURE</code>
R12	Correct note detected an octave off with the expected fundamental present is judged <code>Correct</code>

ID	Test
R13	Helped prompt completes, costs no star, is not counted first-try
R14	Stars thresholds §6.2 for n = 6, 8, 9, 11 at each boundary
R15	Gate question values in [6, 9]; wrong answer does not lock out
R16	Planner offers a repeat at most once per stop; never introduces the new key in warm-up
R17	No audio sample is written to disk (search the storage layer; assert no WAV / PCM writer in release)

13.2 Unit and integration

Onset detector on synthetic signals (silence, one tone, two tones, decaying tone plus new attack, noise burst, speech-like noise); pitch detector on a harmonic series with decay for each of C3–C6; calibration (rejects noisy samples, clears on re-measure); log schema round-trip; export files valid JSON/JSONL; ProgressStore migration from v0.4.0 preferences (do not lose the existing calibration).

Regression harness (debug build only): replays synthetic tones and clips you record yourself through the same pipeline. The release build has no recording code.

13.3 Acceptance gates on the real PSR-F52 (Grand Piano voice) — starting targets

Test	Target
50 clean single presses per key, at 30 cm and 60 cm	≥ 95% named correctly
100 bad inputs (two keys, talking, desk taps, pedal tap)	0 judged as a wrong key; all unclear or ignored
Same key twice, ~1 s apart	≥ 95% counted as two presses

Test	Target
A key held 2 s	< 1% produce a second judgement
Onset to on-screen verdict, median, measured in-app	≤ 150 ms
10 minutes of listening on the target phone	0 dropped frames

Do not claim any latency figure until it has been measured with the log's `onsetToVerdictMs`.

13.4 Child test protocol (before 0.6.0 ships)

Three sittings of about 8 minutes, phone on the stand, adult present but silent for the first 5 minutes. Note: time to first correct press per key; where the child looks (phone or keys); the first moment they get stuck; whether they ask "did it hear me?"; how the session ended. "Did it hear me?" is the key signal: each occurrence means the onset ripple or the unclear message is not doing its job.

14. Releases

Checklists start unchecked. Do not tick anything the code does not yet do.

0.5.0 — "Trust the ears"

Gate: every row of §13.3 passes in Audio Lab.

- ☐ Note names with octaves everywhere (C4, not C); "middle C" beside C4
- ☐ KeyboardGuide (C4–C5) and MiniMap (61 keys); readable labels
- ☐ Audio input: 48k then 44.1k granted-rate, source fallback chain, effects off, mic released on pause/calls/60 s silence, keep-screen-

on

- [] OnsetDetector and PressTracker (one press, one `PressEvent`); overlapping windows; onset timestamps
- [] Judge with stale-press rule and octave logic; `Verdict` and `UnclearReason`
- [] Calibrate: Key dropdown (A2–F6), re-measure clears, mini-map with unhearable keys faded, export
- [] Files tab: Export calibration and session log; Lesson set dropdown
- [] Session log JSONL with `onsetNanos`, `verdictNanos`, `onsetToVerdictMs`
- [] Grown-ups gate: multiplication + press-and-hold
- [] Setup checklist and printable stickers PDF
- [] Tests R1–R6, R11, R12, R15, R17 plus onset/pitch/calibration unit tests
- [] README: how to export the two files

0.6.0 — "Learn"

Gate: §13.4 with the child.

- [] Ladder stops 1–7, LevelPlan generator (R9), unlock (R10), Stars (R7, R8, R14)
- [] Session engine: Warm-up → Learn → Summary/Play; Planner (R16)
- [] Drill screen with onset ripple, distance sentence, persistent amber ring, hint ladder (R13)
- [] Home (with My Five Keys), Level map, Summary, Song screen, six songs (§7), Free Play
- [] Progress tab
- [] Copy deck (§10); accessibility pass (§11)
- [] Tests R7–R10, R13, R14, R16

0.7.0 — "Play"

Gate: hit window tuned from exported logs of real runs.

- ☐ Falling notes judged on `onsetNanos` (window 450 early / 900 late, tunable), missed tile returns once, catches-only counter, stars per §6.3
- ☐ Memory game
- ☐ Map stop 8 becomes playable

After 0.7.0 (only if the child asks for more)

Missing Note, Rhythm, additional sessions from the earlier 26-class syllabus as data, first-class C3–G3 lesson set.

15. Open questions for the owner

1. Confirm **A1** (age 6–8), **A2** (3-activity, ~8 min sessions), **A3** (multiplication gate plus press-and-hold).
2. Confirm the phone position for a sitting (music stand or flat beside the speaker) and whether landscape is wanted; this spec is portrait-only.
3. Confirm the phone model(s) to test on; the acceptance gates depend on the phone's microphone.